

Problem

Evaluate each of the following functions and see if it is periodic. If periodic, find its period.

(a) $f(t) = \cos \pi t + 2 \cos 3\pi t + 3 \cos 5\pi t$

(b) $y(t) = \sin t + 4 \cos 2\pi t$

(c) $g(t) = \sin 3t \cos 4t$

(d) $h(t) = \cos^2 t$

(e) $z(t) = 4.2 \sin(0.4\pi t + 10^\circ) + 0.8 \sin(0.6\pi t + 50^\circ)$

(f) $p(t) = 10$

(g) $q(t) = e^{-\pi t}$

Step-by-step solution

Step 1 of 7

(a) It is given that,

$$f(t) = \cos \pi t + 2 \cos 3\pi t + 3 \cos 5\pi t$$

This function is of the form:

$$f(t) = \cos \omega t + 2 \cos 3\omega t + 3 \cos 5\omega t$$

Clearly $f(t)$ consists of the sum of the fundamental, third and fifth harmonics of cosine function.

Hence $f(t)$ is **periodic**.

The fundamental frequency is,

$$\omega = \pi$$

$$2\pi f = \pi$$

$$2\pi \frac{1}{T} = \pi$$

$$T = 2$$

Thus, the period of the function $f(t)$ is: $T = 2$.

Step 2 of 7

(b) It is given that,

$$y(t) = \sin t + 4 \cos 2\pi t$$

Clearly, $y(t)$ consists of the sine and cosine functions which are periodic.

But the sum of both the functions is not periodic.

Hence $y(t)$ is **not periodic**.

Step 3 of 7

(c) It is given that,

$$g(t) = \sin 3t \cos 4t$$

It is known that,

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

Hence,

$$\begin{aligned} g(t) &= \frac{1}{2} [\sin(3t+4t) + \sin(3t-4t)] \\ &= \frac{1}{2} [\sin(7t) + \sin(-t)] \\ &= -0.5 \sin t + 0.5 \sin 7t \end{aligned}$$

Clearly, $g(t)$ consists of the sum of the fundamental and seventh harmonics of sine function.

Hence $g(t)$ is **periodic**.

The fundamental frequency is,

$$\omega = 1$$

$$2\pi f = 1$$

$$2\pi \frac{1}{T} = 1$$

$$T = 2\pi$$

Thus, the period of the function $g(t)$ is: $T = 2\pi$.

Step 4 of 7

(d) It is given that,

$$h(t) = \cos^2 t$$

It is known that,

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

Hence,

$$\begin{aligned} h(t) &= \frac{1 + \cos 2t}{2} \\ &= 0.5 + 0.5 \cos 2t \end{aligned}$$

Clearly, $h(t)$ consists of the sum of the fundamental and a constant function.

Hence $h(t)$ is **periodic**.

The fundamental frequency is,

$$\omega = 2$$

$$2\pi f = 2$$

$$2\pi \frac{1}{T} = 2$$

$$T = \pi$$

Thus, the period of the function $h(t)$ is: $T = \pi$.

Step 5 of 7

(e) It is given that,

$z(t) = 4.2 \sin(0.4\pi t + 10^\circ) + 0.8 \sin(0.6\pi t + 50^\circ)$ Clearly, from this function, the frequencies are

$$\omega_1 = 0.4\pi \text{ and } \omega_2 = 0.6\pi$$

Take Highest common factor of frequencies.

$$\omega_1 = 2 \times 0.2\pi \text{ and } \omega_2 = 3 \times 0.2\pi$$

Hence the function $z(t)$ is periodic with fundamental frequency $\omega = 0.2\pi$

Calculate the periodicity.

$$\omega = 0.2\pi$$

$$2\pi f = 0.2\pi$$

$$\frac{2\pi}{T} = 0.2\pi$$

$$T = 10$$

Thus, the period of the function $z(t)$ is: $T = 10$.

Step 6 of 7

(f) It is given that,

$$p(t) = 10$$

Since $p(t)$ is a constant, it is **not periodic**.

Step 7 of 7

(g) It is given that,

$$q(t) = e^{-\pi t}$$

The function $q(t)$ must be in the form of $e^{j\omega t}$ to find the periodicity.

But, $q(t) = e^{-\pi t}$

Thus, $q(t)$ is **not periodic**.